



**VIDYAVARDHINI'S COLLEGE OF ENGINEERING AND TECHNOLOGY**  
(NBA & NAAC Accredited)  
(AUTONOMOUS INSTITUTE AFFILIATED TO UNIVERSITY OF MUMBAI)



# OSCILLATIONS'26

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# Oscillations '26

## Technical Paper Presentation

Title : **Predictive Degradation and Failure  
Detection Intelligence System**

Track No : 02

Group No: 01

Group Member: Sanchit Gaikwad  
Aditya Ghanekar  
Omkar Hmabire  
Mehul HIngu



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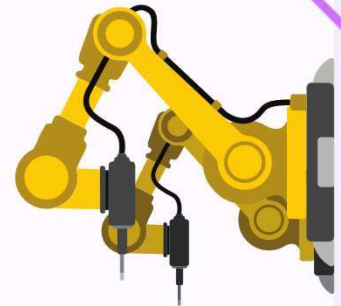


## INTRODUCTION

Machines in industries and workshops often fail unexpectedly, leading to production losses, increased repair costs, and safety hazards. Traditional approaches—either reactive maintenance (waiting for failure) or scheduled maintenance—are inefficient and unable to predict faults in advance.

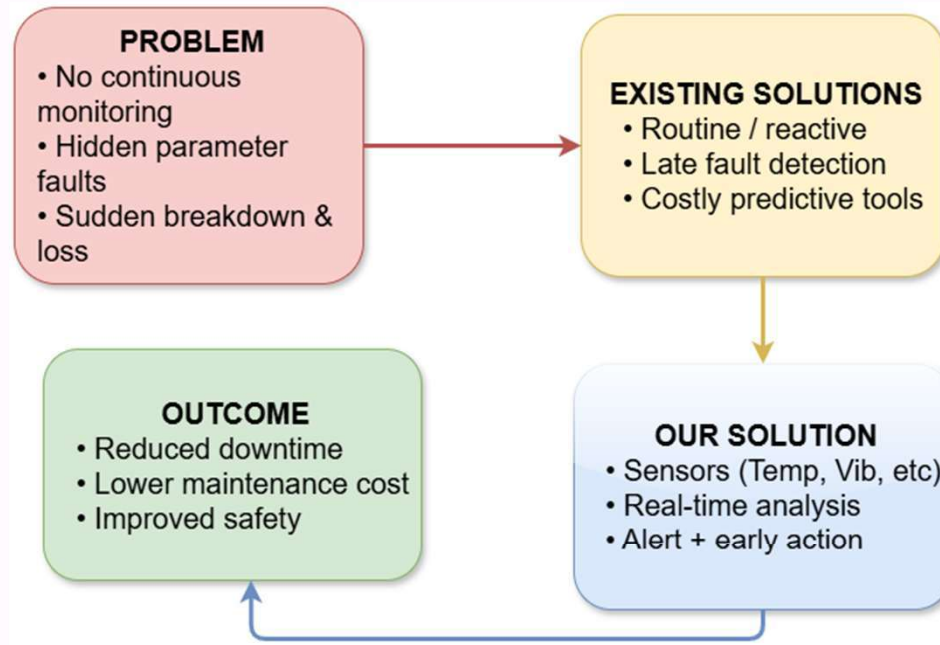
Existing monitoring systems used in large industries are expensive and require skilled operators, making them unsuitable for small industries, workshops, and college labs.

To solve this, we developed a low-cost, portable Machine Health Monitor using Arduino Nano and sensors to detect abnormal conditions early and alert users, preventing breakdowns and improving safety.





## ABSTRACT



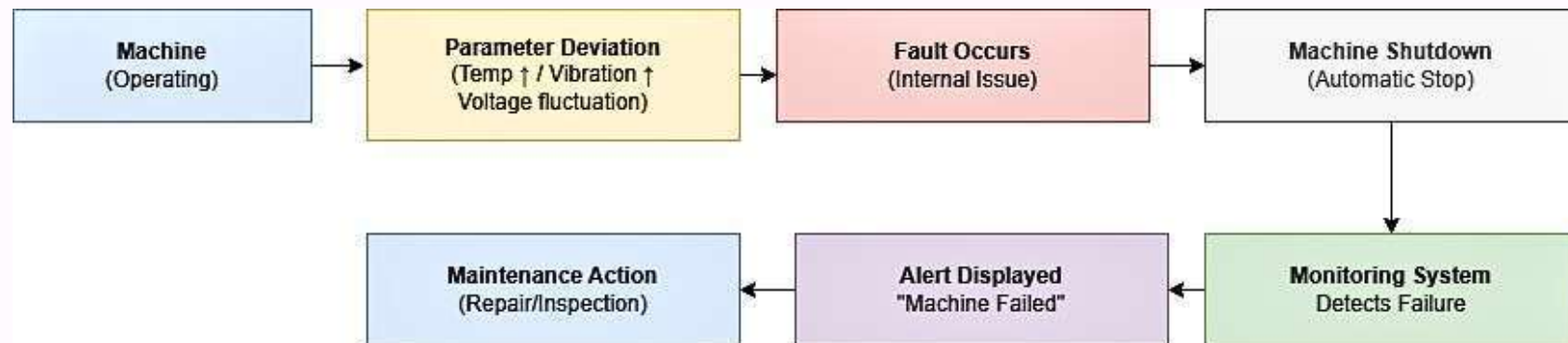


## LITERATURE SURVEY

| Title of Paper                                           | Year | Key Findings                                                                                  | Gaps                                                   |
|----------------------------------------------------------|------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------|
| IoT-Based Machine Health Monitoring System               | 2020 | Proposed real-time monitoring using sensors and cloud connectivity for predictive maintenance | High cost and dependency on internet connectivity      |
| Predictive Maintenance using Machine Learning Techniques | 2021 | Used ML algorithms like SVM and Random Forest for fault prediction with improved accuracy     | Requires large dataset and high computational power    |
| Multi-Sensor Data Fusion for Fault Detection             | 2019 | Combined multiple sensor data to improve fault detection reliability                          | Complex implementation and increased system cost       |
| Vibration-Based Condition Monitoring of Machines         | 2018 | Identified faults using vibration analysis as a primary parameter                             | Limited to mechanical faults, ignores other parameters |



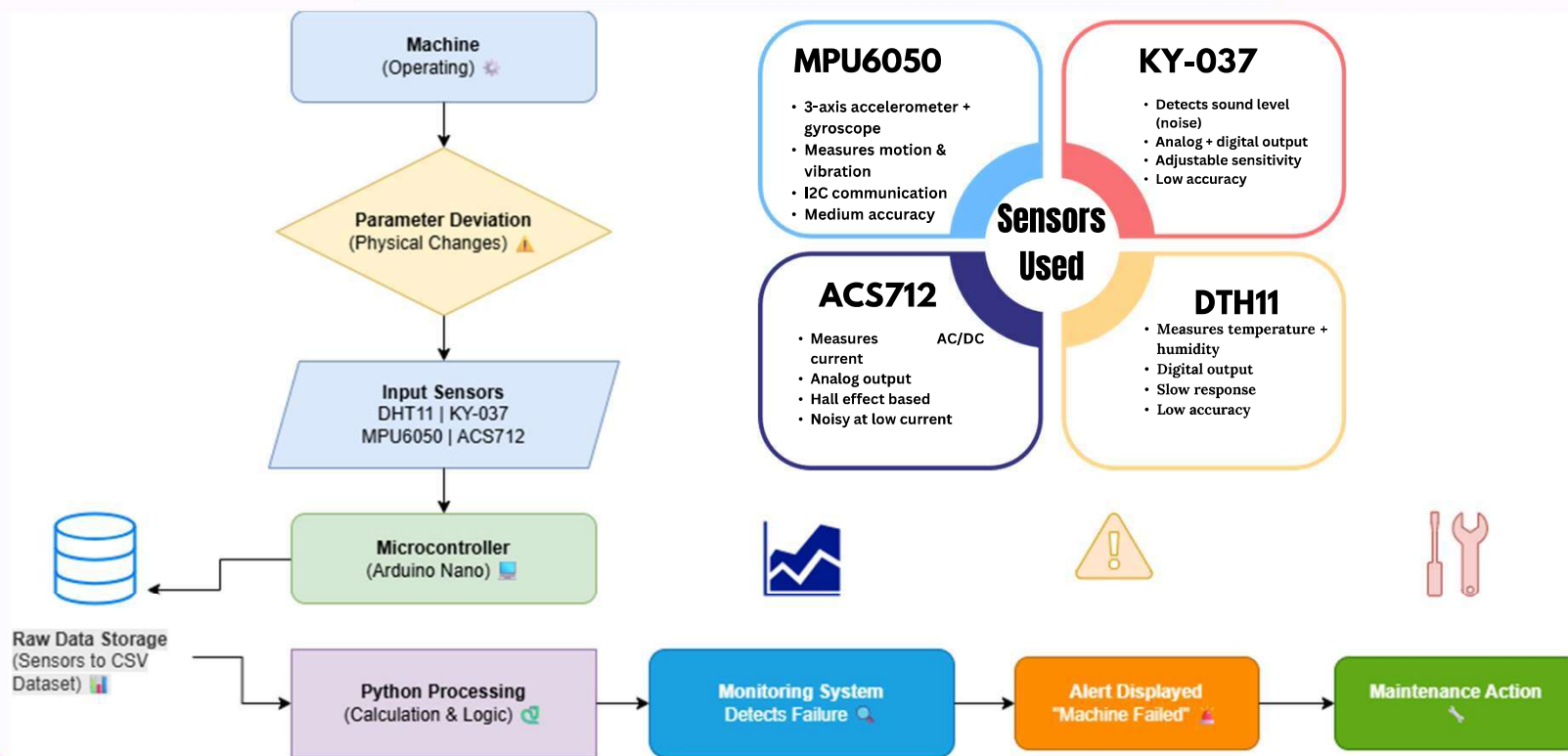
## PROBLEM DEFINITION





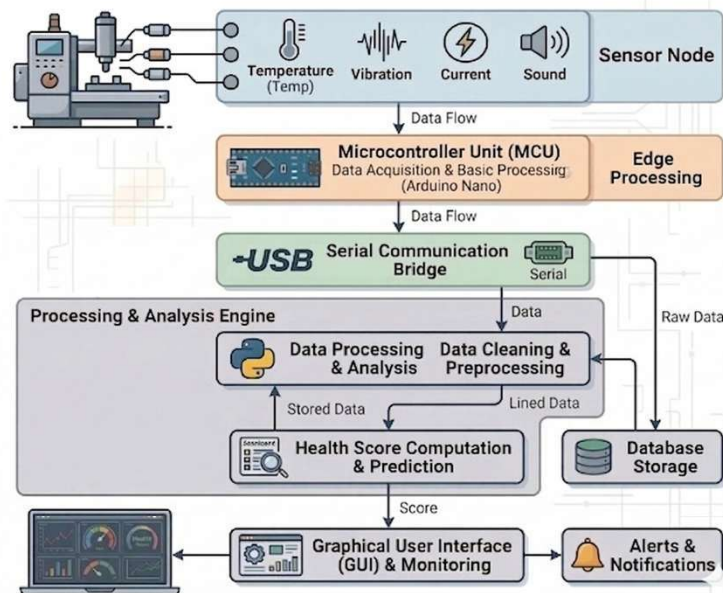


## METHODOLOGY





## System Design

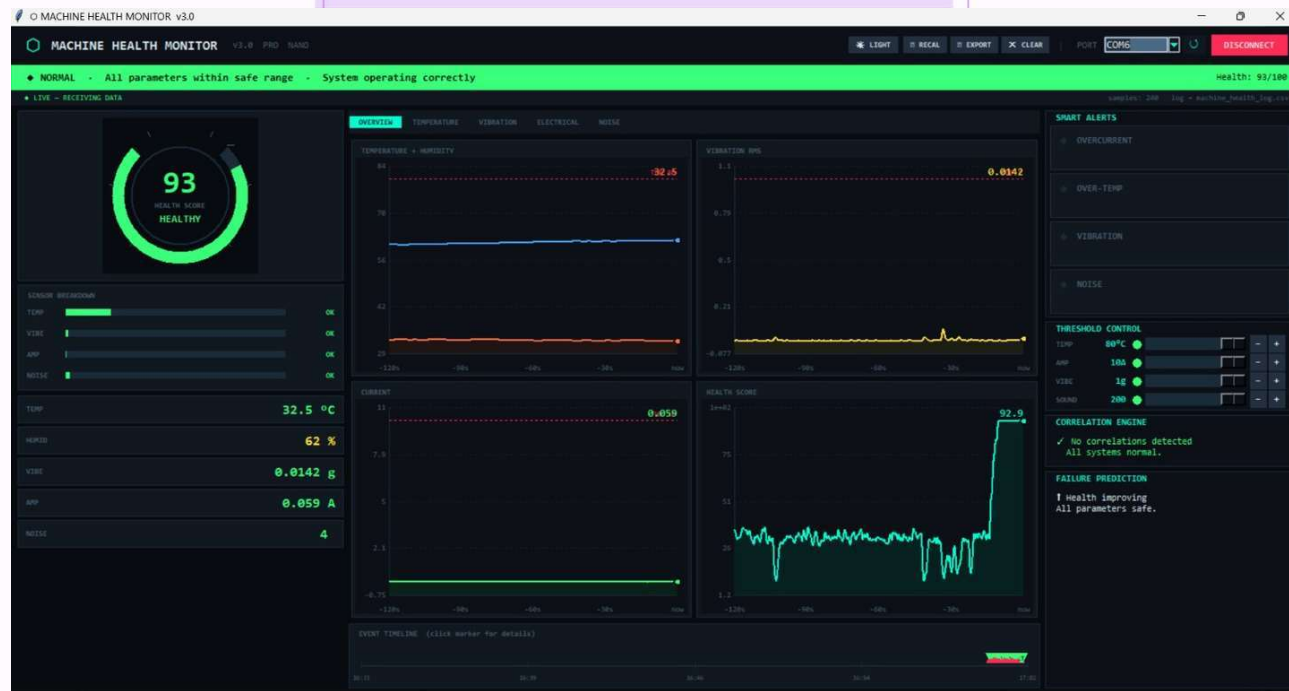


- **Multisensory Input:** Sensors (DHT11, MPU6050, KY-037, ACS712) capture temperature, vibration, sound, and current.
- **Edge Processing:** An Arduino Nano performs initial data acquisition and basic filtering.
- **Data Bridge:** Information is sent via USB Serial Communication to a central processing unit.
- **Python Engine:** A Python script executes signal processing, cleaning, and feature extraction.
- **Health Scoring:** The system computes a Health Score to predict potential internal faults.
- **Monitoring & Alerts:** Results are displayed on a GUI dashboard, triggering alerts or automatic shutdowns if failure is detected.



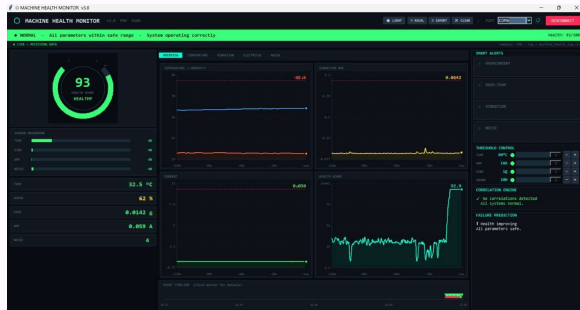


## RESULT AND DISCUSSION





## RESULT AND DISCUSSION

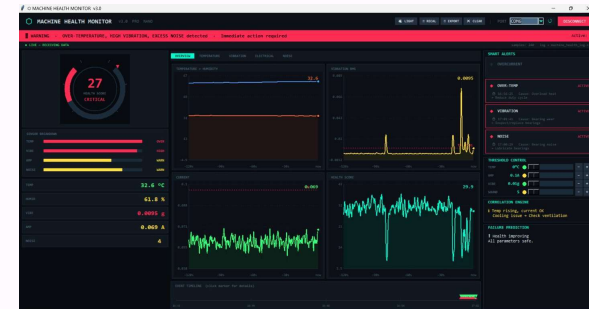


### Healthy System

- All monitored parameters operate within normal and stable conditions
- No abnormal patterns or deviations observed in sensor data
- System runs smoothly without triggering any fault indications

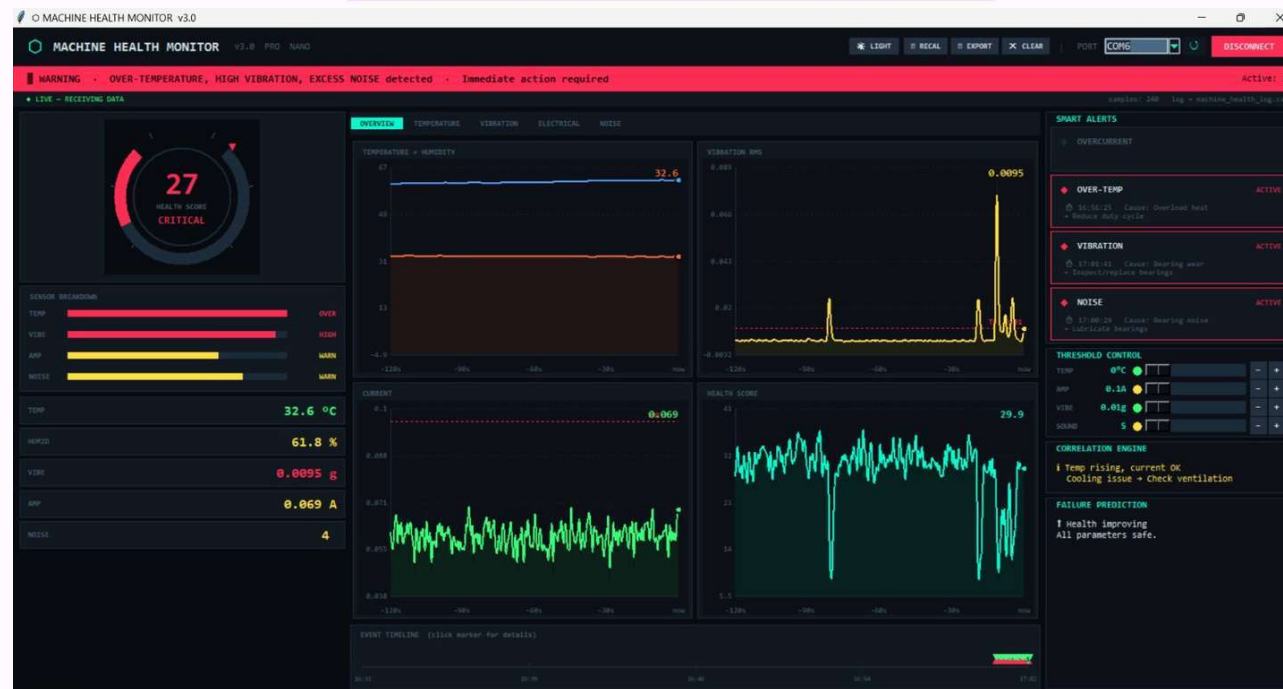
### Critical System

- Multiple parameters show abnormal behavior beyond acceptable limits
- System exhibits unstable and rapidly degrading performance
- Fault conditions are detected, triggering continuous alerts





## RESULT AND DISCUSSION



[Project video](#)





## CONCLUSION

Hence, we Conclude that A real-time machine health monitoring and failure prediction system using multi-sensor data fusion. By integrating temperature, vibration, current, and sound sensing, the system continuously evaluates machine condition. A weighted health score and linear regression-based prediction model enable early detection of faults such as overheating, excessive vibration, electrical instability, and acoustic anomalies. The solution is cost-effective, scalable, and suitable for small-scale industries and academic research, providing a solid foundation for predictive maintenance that can be further improved with advanced signal processing and machine learning.

Integration of advanced machine learning models for improved prediction accuracy

IoT-based cloud connectivity for remote monitoring and data analytics

**Future**

**Scope**

Use of advanced signal processing (FFT, wavelet) for precise fault detection

Addition of automated alert and control systems for autonomous operation



## REFERENCES

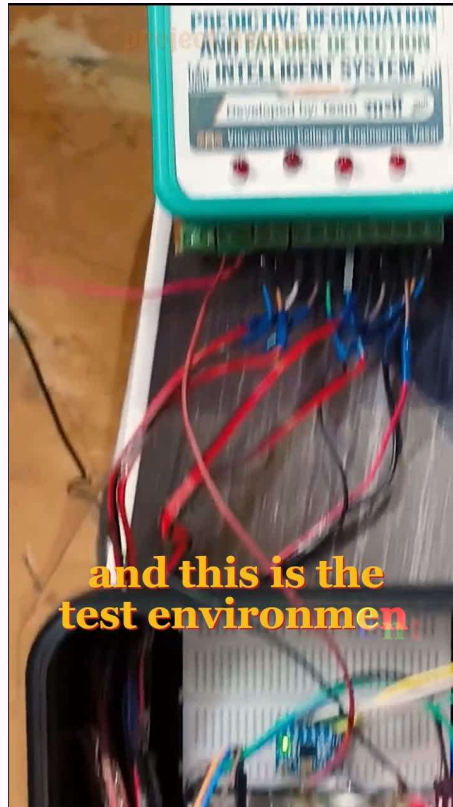
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## VIDEO



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